DSC 257R: Unsupervised learning

Homework 6

Multivariate Gaussians and Gaussian processes

1. Consider the spherical Gaussian $N(0, I_d)$ in d-dimensional space.

   (a) What is the maximum density achieved by this distribution? At what point is it achieved?

   (b) In very rough terms, what happens to the maximum density as d grows? (This helps explain why for high-dimensional data we typically work with the logarithm of the density, in order to avoid numerical underflow errors.)

2. *Length concentration.* Write a procedure that draws 10000 points $X_1, \ldots, X_{10000}$ from the spherical Gaussian $N(0, I_d)$ and plots a histogram of their (squared) lengths $\|X_i\|^2$.

   (a) Do this for $d = 10$. What seems to be the typical (squared) length?

   (b) Repeat for $d = 100$. Is the histogram more concentrated than in (a), or is it less concentrated?

3. Suppose $X \in \mathbb{R}^2$ has a Gaussian distribution with mean $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and covariance matrix $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$. What is the distribution of the projection of X onto direction $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

4. We have three sensors (X_1, X_2, X_3) whose joint distribution is Gaussian with mean zero and covariance matrix
$$\begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

   (a) What is the distribution of sensor X_3?

   (b) What is the joint distribution of sensors X_1, X_2?

   (c) Suppose we observe $X_2 = 2$ and $X_3 = 1$. What is the resulting conditional distribution of X_1?

5. *Modeling temperature data with Gaussian processes.* In this question, we will explore modeling of geospatial data via Gaussian processes. Begin by downloading the files `gptrain.csv` and `gptest.csv` from the course webpage. Each row in each file corresponds to a temperature reading at a given weather station in Brazil at 2pm on January 1, 2021. The first column gives the `latitude` of the station, the second the `longitude`, and the final column is the `temperature` reading in degrees Celsius. We will fit a simple Gaussian Process model on `gptrain.csv` and use it to infer temperatures at other locations at that same date and time.

   We will fit a Gaussian process with covariance function
$$k(x, x') = \exp\left(\frac{-\|x - x'\|}{\sigma}\right),$$

and mean function $m(x) = 20$ (degrees Celsius). Here $\|x - x'\|$ denotes Euclidean distance between raw latitude and longitude coordinates.

Denote the locations in `gptrain.csv` by X_{tr} and the temperatures at those locations by y_{tr}; define X_{te}, y_{te} analogously for `gptest.csv`. Thus X_{tr} is a 93×2 matrix while y_{tr} is a 93-dimensional vector.

(a) The joint distribution of training and test responses can be written as

$$\begin{bmatrix} y_{\text{tr}} \\ y_{\text{te}} \end{bmatrix} \sim N\left(\begin{bmatrix} 20 \\ \vdots \\ 20 \end{bmatrix}, \quad \begin{bmatrix} K_{\text{tr}} & K_{\text{tr,te}} \\ K_{\text{te,tr}} & K_{\text{te}} \end{bmatrix} \right),$$

where the block matrix

$$\begin{bmatrix} K_{\text{tr}} & K_{\text{tr,te}} \\ K_{\text{te,tr}} & K_{\text{te}} \end{bmatrix}$$

is the matrix arising from all pairwise evaluations of the covariance function $k(\cdot)$ over all training and testing locations, and its diagonal components K_{tr} and K_{te} are the matrices arising from all pairwise covariances within the training and test sets, respectively.

Explain very briefly how the definition of Gaussian process allows us to come to this conclusion.

(b) Define

$$m_{\text{te}} := \begin{bmatrix} 20 \\ \vdots \\ 20 \end{bmatrix} \in \mathbb{R}^{11},$$

since there are 11 locations in the test set. Using properties of the Gaussian, it can be shown that the distribution of test responses given training locations, training responses, and test locations, follows

$$\begin{aligned} & y_{\text{te}} \mid y_{\text{tr}}, X_{\text{te}}, X_{\text{tr}} \\ & \sim m_{\text{te}} + N\left(K_{\text{te,tr}} \cdot K_{\text{tr}}^{-1} \left(y_{\text{tr}} - m_{\text{tr}} \right), \quad K_{\text{te}} - K_{\text{te,tr}} K_{\text{tr}}^{-1} K_{\text{tr,te}} \right), \end{aligned}$$

In Bayesian language, this is the posterior predictive distribution.

- Using X_{tr}, X_{te}, and y_{tr}, compute the posterior mean of y_{te} via matrix multiplication, using $\sigma = 1.5$ in the covariance function. Report the mean squared error between your computed posterior mean and the true values found in y_{te}.
- Report the mean squared error that you would have obtained by simply predicting the average value in y_{tr} for all locations.

(c) The diagonal of the posterior predictive matrix

$$K_{\text{te}} - K_{\text{te,tr}} K_{\text{tr}}^{-1} K_{\text{tr,te}}$$

gives the conditional variances of the test responses given X_{tr}, X_{te}, and y_{tr}.

- Create a rich grid (at least 5000 points) of test latitudes and longitudes within the range found in the training set. Make a plot showing the predicted (posterior mean) temperature at *all points in the grid*. You might find the function `matplotlib.pyplot.imshow` helpful for this.

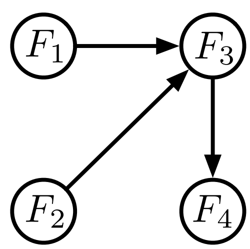
- Make another plot showing the standard deviation of the prediction at *all points in the grid*. Plot the locations of the stations found in the training set as well.
- What do you notice about the relationship of the standard deviation of the prediction and the distance of that prediction to a training point?

Bayes nets and autoregressive models

6. Four friends—F_1, F_2, F_3, F_4—like going to the movies. When a new movie comes out, they behave according to the following scheme:

   - F_1 sees the movie with probability 0.5.
   - F_2 sees the movie with probability 0.7, independently.
   - If either of F_1 or F_2 (or both) see the movie, F_3 goes to see it with probability 0.8; if neither F_1 nor F_2 sees the movie, F_3 sees it with probability 0.3.
   - If F_3 sees the movie, then F_4 goes to see it with probability 0.9. If F_3 doesn't see the movie, then F_4 sees it with probability 0.5.

   Let $F_1, F_2, F_3, F_4 \in \{0, 1\}$ denote who sees the movie. The dependencies between these four random variables can be captured by the following directed graph:

   (a) Give the conditional probability tables for this Bayes net.

   (b) What is the probability of the configuration $F_1 = F_2 = F_3 = 1, F_4 = 0$?

7. *Non-uniqueness of underlying DAG.* Suppose random variables X and Y take values in $\{0, 1\}$. Does the joint distribution of (X, Y) necessarily factor over the DAG on the left below? What about the DAG on the right?

8. *Genes, diseases, and outcomes.* There is a fictitious disease (D) that can cause paralysis (P). Whether or not an individual gets this disease depends in large part on two genes, G_1 and G_2. Here is a summary:

   - Each of these genes G_1, G_2 can either be normal (also known as *wild-type*), represented by $G_i = 0$, or *mutant*, represented by $G_i = 1$.
   - For a person with $G_1 = G_2 = 0$ (both genes wild-type) the probability of acquiring the disease is zero. If G_1 is mutant and G_2 is wild-type, the probability is 0.2. If G_1 is wild-type and G_2 is mutant, it is 0.1. And if both genes are mutant, then the disease probability is 0.5.

- For an individual with the disease, the probability of paralysis is 0.5. Without the disease, there is still a small probability of paralysis (from other causes): 0.01.
- G_1 is mutant with probability 0.1 and G_2 is mutant with probability 0.2; these are independent.

(a) Draw a suitable Bayes net over the variables G_1, G_2, D, P, and give all relevant conditional probability tables. Let $D = 1$ if the disease is present and $D = 0$ if it isn't; likewise with paralysis.

(b) What is the probability of disease ($D = 1$)?

(c) What is the probability of paralysis ($P = 1$)?

(d) We observe that a person has paralysis. What is the probability that they have the disease?

(e) We observe that a person has the disease ($D = 1$). What is the probability that they have the mutant form of gene G_1?

9. *A character-level RNN for text generation.* Download the file `rnn-gen.zip` from the course webpage; it contains a Jupyter notebook, `text-gen-rnn.ipynb`, as well as data and helper files. The notebook is adapted from an excellent blog post by Andrej Karpathy, *The unreasonable effectiveness of recurrent neural networks*. It fits an RNN (more precisely, an LSTM) to the text of Shakespeare's plays and then uses this to generate more text, one character at a time.

   Work through the notebook and try to understand what is going on. There are just three small questions to answer.

   (a) How many characters are we considering, and what is the integer code for 'A'?

   (b) What is the length of the corpus in characters?

   (c) The input and output size for the RNN are set to 66. Why is this?